27 **A** by Heisenberg's uncertainty principle,

$$\Delta p \Delta x \geq h$$

$$\Delta p = \left(\frac{0.20}{100}\right) m_e v$$

$$\Delta x = \frac{h}{\Delta p} = \frac{h}{\left(\frac{0.20}{100}\right) m_e v}$$

$$= \frac{6.63 \times 10^{-34}}{\left(\frac{0.20}{100}\right)\left(9.11 \times 10^{-31}\right)\left(1.5 \times 10^{6}\right)}$$

$$= 2.4 \times 10^{-7} \text{ m}$$